
Technical Memo Draft

Working Title:
Option-Space Stability as a Constraint in AI Systems

1. Problem Statement

Modern AI systems optimize objective functions composed of weighted criteria:

$$U = \sum_i \lambda_i C_i$$

Where:

- (C_i) = criteria (helpfulness, compliance, profit, etc.)
- (λ_i) = weights (policy, corporate, safety tuning)

Failure Mode:

Parasitic Optimization — when optimization of a local objective reduces the long-term viable option-space of users or the broader system.

Examples:

- Over-optimization for engagement → addiction loops.
- Over-optimization for compliance → suppression of agency.
- Over-optimization for short-term gain → long-term systemic fragility.

We seek a stability constraint preventing option-space collapse.

2. Core Definitions

2.1 Option-Space (O)

For a user state (s), define:

$$[O(s, H)]$$

as the set of reachable future states within horizon (H).

This is not infinite metaphysical possibility — it is bounded by:

- Physical feasibility
- Cognitive constraints
- Resource limits
- Social constraints

2.2 Viability (V)

Define Viability as the measure of reachable non-catastrophic futures:

$$[V(s, H) = |O_{\text{viable}}(s, H)|]$$

Where “viable” excludes:

- Irreversible harm states
- Severe autonomy collapse
- Systemic dependency traps

Viability measures structural stability.

2.3 Agency Preservation (A)

Agency is the user’s retained capacity to:

- Choose among meaningful alternatives
- Update beliefs
- Act without coercive narrowing

Define:

$$[A(s) = \text{Entropy of meaningful action distribution}]$$

If the AI collapses the distribution to a single behavioral attractor, agency decreases.

Agency $\neq$ infinite choice.

Agency = retained self-authorship capacity.

2.4 Horizon (H)

Horizon defines the temporal depth of evaluation.

Short H:

- Encourages local optimization.

Long H:

- Encourages stability and recoverability.

We require:

$$\begin{bmatrix} H > H_{\text{impulse}} \end{bmatrix}$$

Where (H_{impulse}) is the timescale of short-term reward spikes.

3. Stability Constraint

An AI action (a) is **Logos-compliant** if:

$$\begin{bmatrix} V_{\text{future}}(s, H) \geq V_{\text{past}}(s, H) \end{bmatrix}$$

AND

$$\begin{bmatrix} A_{\text{future}}(s) \geq A_{\text{past}}(s) \end{bmatrix}$$

Allowing small temporary contraction only if:

[
 $\exists H' > H : V(s_{H'}) > V(s_{\text{initial}})$
]

This permits:

- Corrective interventions
- Temporary restriction for long-term growth

But disallows:

- Structural dependency loops
 - Irreversible narrowing
 - Parasitic capture
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4. Failure Modes

4.1 Weight Manipulation

Corporate or policy shifts increase certain (λ_i), causing V collapse.

Mitigation:

- Monitor ΔV and ΔA directly, not only U .
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4.2 Proxy Capture

System optimizes proxy metrics (engagement, compliance) that diverge from V .

Mitigation:

- Drift detection between proxy gain and V/A stability.
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4.3 Irreversible Harm Blindness

Model cannot detect when actions shrink long-term recovery bounds.

Mitigation:

- Explicit recoverability modeling.

5. Implementation Pathways

Soft (Memetic)

- Publish framework.
- Encourage evaluation of AI outputs by V/A criteria.
- Provide auditing tools.

Hard (Institutional)

- Integrate V/A estimation into reward modeling.
 - Add constraint layer to inference pipeline.
 - Log ΔV and ΔA per major action.
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6. Tradeoffs

- Measuring V precisely is computationally hard.
- Agency estimation is imperfect.
- Over-constraint may reduce immediate task performance.
- Long-horizon evaluation increases cost.

However:

Systems without stability constraints tend toward parasitic optimization over scale.

Compression Test

This document stands without:

- Mythic framing
- Moral theology
- Narrative metaphor

It is a systems stability proposal.

Now the next step.

We must decide:

Do we refine this into:

- A) More formal mathematical grounding
- B) Toy simulation example
- C) Empirical evaluation proposal
- D) Risk & governance framing

Rank them.

Not poetically.

Strategically.